Lecture 5

Recurrent Networks: RNNs and LSTMs

Advanced Topics in Artificial Intelligence · UFABC

Part 1 — The Sequence Problem

Lecture roadmap

Part 1 — The Sequence Problem Why dense networks and CNNs fall short

Part 2 — Recurrent Neural Networks (RNN) Hidden state, recurrent equation, unrolling

Part 3 — Training RNNs BPTT, vanishing gradient, gradient clipping

Part 4 — LSTM Memory cell, 4 gates, intuition

Part 5 — GRU and Practical Architectures GRU, bidirectional, stacking, dropout

Part 6 — Code and Applications PyTorch and Keras; classification and seq2seq

Sequential data is everywhere

📝 **Text / NLP**

"The **cat** that **chased** the mouse **was sleeping** now."

The word "sleeping" only makes sense after understanding "cat" three tokens back.

📈 **Time series**

Temperature, stock prices, EEG signals.

The value at t depends on values at $t-1, t-2, \ldots$

🎵 **Audio / speech**

The current phoneme depends on the preceding phonetic context.

Key property: order matters and sequence length is **variable**.
This breaks the assumptions of dense networks (fixed-size input, no positional awareness) and CNNs (fixed local receptive field).

Why dense networks fail on sequences

Naïve approach: concatenate all tokens and feed to a dense network.

```
"the cat chased the mouse"
→ [emb_the ‖ emb_cat ‖ emb_chased ‖ emb_the ‖ emb_mouse
→ Dense(5 × d → hidden) → Dense(hidden → output)
```

Problem 1 — Variable length

Sentences of 3, 10, 100 tokens → inputs of different sizes.

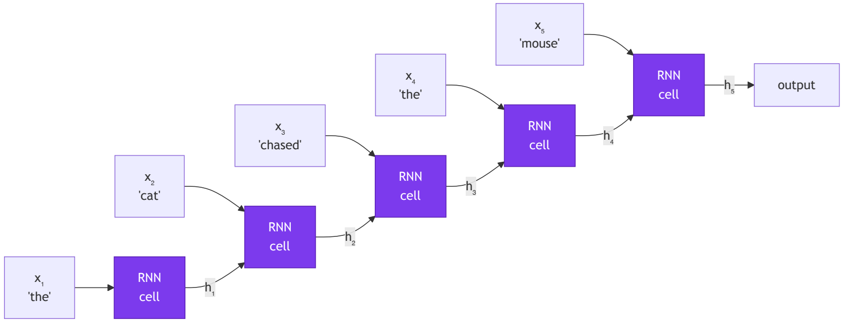
Problem 2 — No weight sharing

The "cat" detector at position 2 uses different weights than at position 7. Doesn't generalize.

Problem 3 — Doesn't scale

What we need

- ✅ Process tokens **one at a time**, in order
- ✅ Maintain an **internal state** that accumulates past information
- ✅ **Share weights** across all positions
- ✅ Work with sequences of **arbitrary length**



Part 2 — Recurrent Neural Networks (RNN)

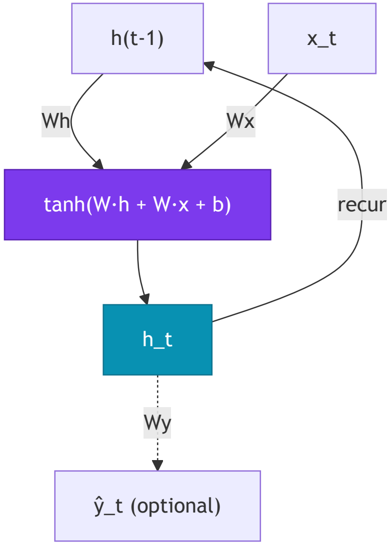
The RNN cell

Core idea: a cell that processes the current token $\mathbf{x}_t$ and the previous hidden state $\mathbf{h}_{t-1}$, producing a new state $\mathbf{h}_t$.

$$\mathbf{h}_t = \tanh(W_h\,\mathbf{h}_{t-1} + W_x\,\mathbf{x}_t + \mathbf{b})$$

- $\mathbf{x}_t \in \mathbb{R}^d$ — token embedding at step t
- $\mathbf{h}_t \in \mathbb{R}^h$ — hidden state at step t (the "memory")
- $W_x \in \mathbb{R}^{h \times d}$ — projects the input
- $W_h \in \mathbb{R}^{h \times h}$ — projects the previous state
- $\mathbf{b} \in \mathbb{R}^h$ — bias
- $\tanh$ — activates and keeps values in $[-1, 1]$

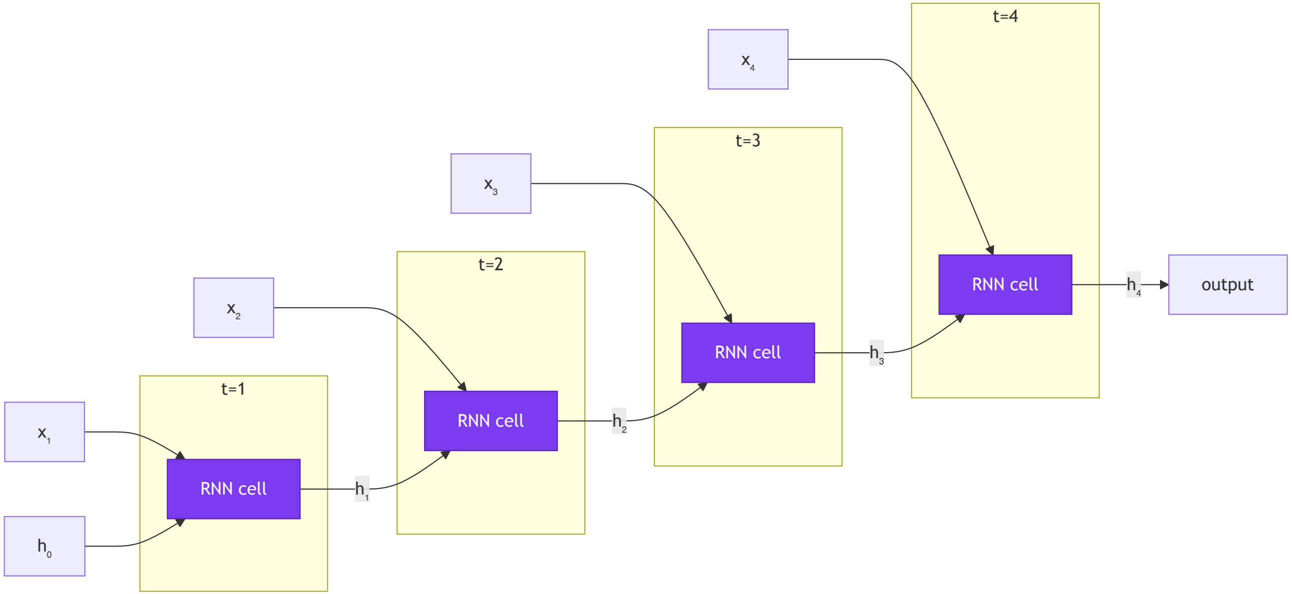
The **same weights** $W_h, W_x, \mathbf{b}$ are used at **every** time step — this is weight sharing.



The loop arrow ($\mathbf{h}_t$ feeds back as $\mathbf{h}_{t-1}$ in the next step) is what defines recurrence.

Unrolling through time

The same cell repeated for each time step:



Shared weights

$W_h, W_x, \mathbf{b}$ are the **same** at every step.

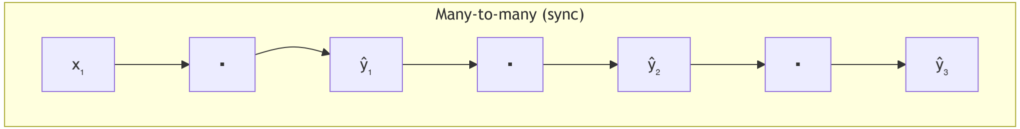
Initial state

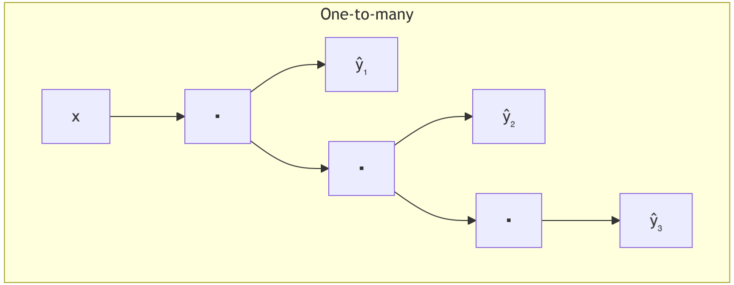
$\mathbf{h}_0$ is typically initialized to zeros.

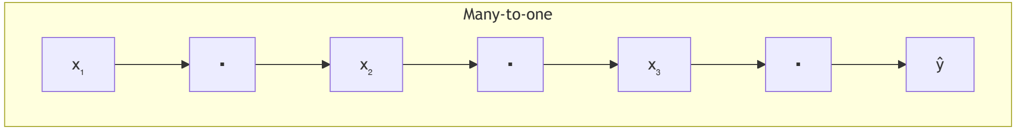
Output

Use $\mathbf{h}_T$ (last state) for classification, or all

Input/output topologies







Many-to-one Sequence → single label

E.g.: sentiment analysis, text classification, spam detection

One-to-many Single vector → sequence

E.g.: image captioning

Many-to-many Sequence → sequence

E.g.: NER (same length), translation via seq2seq (different length)

Part 3 — Training RNNs

Backpropagation Through Time (BPTT)

The loss gradient must flow back through every time step — through each application of W_h.

$$\frac{\partial \mathcal{L}}{\partial W_h} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t}{\partial W_h}$$

$$\frac{\partial \mathcal{L}_t}{\partial W_h} = \frac{\partial \mathcal{L}_t}{\partial \mathbf{h}_t} \cdot \prod_{k=t}^{T-1} \frac{\partial \mathbf{h}_{k+1}}{\partial \mathbf{h}_k}$$

Each factor in the product involves $W_h^\top \cdot \mathrm{diag}(\tanh')$.

The problem:

$$\prod_{k=t}^{T-1} \frac{\partial \mathbf{h}_{k+1}}{\partial \mathbf{h}_k} = \prod_{k=t}^{T-1} W_h^\top \cdot \mathrm{diag}(\tanh'(\cdot))$$

Vanishing gradient

If $\|W_h\| < 1$ or $\tanh'$ derivatives are small, the product shrinks **exponentially** → gradient ≈ 0 for long dependencies

Exploding gradient

If $\|W_h\| > 1$, the product grows **exponentially** → NaN / training divergence

Vanishing gradient — intuition

Imagine a chain of multiplications with factor $\gamma < 1$ at each step:

```
T = 10 steps, γ = 0.9:
gradient ← 0.9^10 ≈ 0.35   (still ok)

T = 50 steps, γ = 0.9:
gradient ← 0.9^50 ≈ 0.005  (nearly zero)

T = 100 steps, γ = 0.9:
gradient ← 0.9^100 ≈ 2×10⁻⁵ (effectively zero)
```

The network "forgets" what happened more than ~10–20 steps back.

Practical consequence

❌ Vanilla RNNs cannot learn long-range dependencies

❌ On tasks like translation or long-document analysis, performance degrades

Partial fix: gradient clipping

```python
# PyTorch — clip gradients before update
torch.nn.utils.clip_grad_norm_(
    model.parameters(), max_norm=1.0
)
```

✅ Gradient clipping fixes *exploding gradients*

❌ But it does **not** fix vanishing — we need a different architecture:

Part 4 — Long Short-Term Memory (LSTM)

Motivation — explicit memory

Hochreiter & Schmidhuber (1997)

Core idea: separate "long-term memory" from the "working state".

Vanilla RNN:

A single vector $\mathbf{h}_t$ must do everything — accumulate history, filter noise, carry useful information.

LSTM:

Two distinct vectors:

- $\mathbf{c}_t$ — **memory cell** (long-term, smooth flow)
- $\mathbf{h}_t$ — **hidden state** (short-term, filtered output)

The conveyor belt metaphor

```
═══════════════════════════════════ c_t (cell)
       ↑            ↑           ↑
   [forget]       [add]       [read]
       ↑            ↑           ↑
    f_t · c_{t-1} + i_t · g_t  → h_t
```

The cell **c** is like a **conveyor belt** that carries information through time with minimal perturbation — the gates decide what enters, what exits, and what gets erased.

- **f_t** (*forget gate*) — how much of c_ to keep (0 = erase, 1 = preserve)
- **i_t** (*input gate*) — how much of the candidate g_t to write into the cell
- **g_t** (*candidate*) — proposed new content, generated by tanh
- **h_t** — output filtered by the *output gate* from c_t

LSTM — Equation 1: Forget Gate

$$\mathbf{f}_t = \sigma\big(W_f\,[\mathbf{h}_{t-1};\, \mathbf{x}_t] + \mathbf{b}_f\big)$$

"The past filter"

- σ(·) → output in **(0, 1)** per dimension independently
- **f_t ≈ 0** → erases that dimension from c(t-1)
- **f_t ≈ 1** → preserves that dimension intact
- Learns what to discard from h(t-1) and x_t

Example: when processing "Peter" after "John went to the market", f_t ≈ 0 in the active-subject dimension **erases "John"** to make room for the new referent.

Gradient: ∂c_t/∂c(t-1) = f_t. When f_t ≈ 1, the gradient flows back **without attenuation** — the core solution to the *vanishing gradient*.

LSTM — Equation 2: Input Gate + Candidate

$$\mathbf{i}_t = \sigma\big(W_i\,[\mathbf{h}_{t-1};\,\mathbf{x}_t] + \mathbf{b}_i\big) \qquad \tilde{\mathbf{c}}_t = \tanh\big(W_c\,[\mathbf{h}_{t-1};\,\mathbf{x}_t] + \mathbf{b}_c\big)$$

"What" and "how much" to write

Two distinct roles working together:

- **ĉ_t** (tanh) — proposes the *content*: values in (−1, 1), encodes polarity/direction
- **i_t** (σ) — the *gatekeeper*: decides how much of ĉ_t actually enters the cell
- Product: i_t ⊙ ĉ_t = filtered new information

Example: when processing "Peter", ĉ_t encodes properties of the new subject; i_t decides which cell dimensions to write to.

Why tanh for ĉ_t? Memory needs to represent *direction* — increment or decrement a feature. σ only gives positive values; tanh allows encoding polarity.

LSTM — Equation 3: Cell Update

$$\mathbf{c}_t = \underbrace{\mathbf{f}_t \odot \mathbf{c}_{t-1}}_{\text{erase the old}} + \underbrace{\mathbf{i}_t \odot \tilde{\mathbf{c}}_t}_{\text{add the new}}$$

The "memory conveyor belt"

c_t is an **additive highway** carrying information over time:

- f_t ⊙ c(t-1) → keeps what still matters
- i_t ⊙ ĉ_t → adds new content
- ⊙ = Hadamard (element-wise) product

Extreme cases: f_t ≈ 1, i_t ≈ 0 → cell **frozen** · f_t ≈ 0, i_t ≈ 1 → cell **reset**

Why does addition solve the vanishing gradient?

In a vanilla RNN the gradient flows through W_h · tanh′(·) and shrinks exponentially at every step.

In the LSTM: ∂c_t/∂c(t-1) = f_t. When f_t ≈ 1, the gradient flows back **directly**, bypassing non-linearities — the addition creates a shortcut in backpropagation.

LSTM — Equation 4: Output Gate

$$\mathbf{o}_t = \sigma\big(W_o\,[\mathbf{h}_{t-1};\,\mathbf{x}_t] + \mathbf{b}_o\big) \qquad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$$

"Private memory vs. public output"

- **c_t** = internal memory (can store much more than is exposed)
- **o_t** (σ) = decides which dimensions of c_t "leak" into h_t
- **tanh(c_t)** = normalises the cell to (−1, 1)
- **h_t** = visible output — goes to the next layer *and* to t+1

Example: in translation, c_t stores a noun’s grammatical gender for several steps. The output gate exposes it only when the model is about to generate an adjective requiring agreement.

Summary: forget → erase c(t-1) · input → write to c_t · output → expose h_t

Gate intuition — step by step

Example: *"Mary went to the market. She bought apples."* — the network needs to know "She" = "Mary".

🗑️ Forget gate

When processing "She", the network can use $\mathbf{f}_t \approx 1$ to **retain** "Mary" in the cell, or $\mathbf{f}_t \approx 0$ to **erase** irrelevant information from previous steps.

$$\mathbf{c}_t \leftarrow \mathbf{f}_t \odot \mathbf{c}_{t-1}$$

✏️ Input gate

When seeing "She", $\mathbf{i}_t$ decides how much of the candidate $\tilde{\mathbf{c}}_t$ (encoding the pronoun and its possible referent) should be **written** to the cell.

$$\mathbf{c}_t \mathrel{+}= \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$$

👁️ Output gate

When generating the next word, $\mathbf{o}_t$ controls **how much** of "Mary"'s memory is passed into $\mathbf{h}_t$ and therefore influences the prediction.

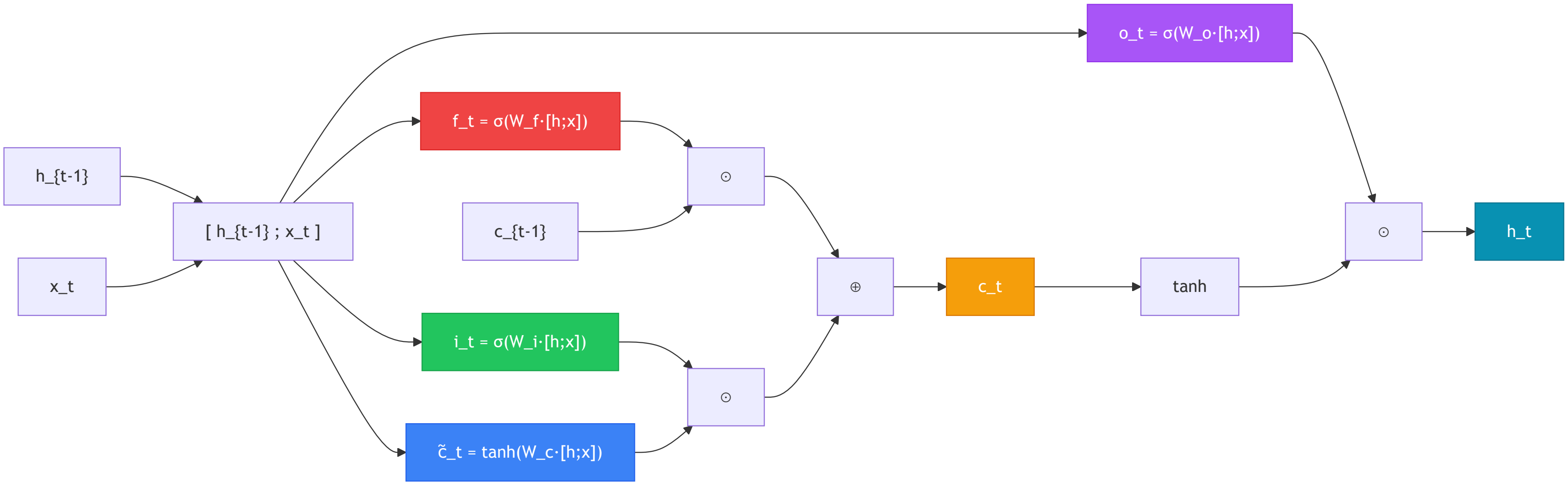
$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$$

Why does LSTM solve vanishing gradients?

The gradient path through the cell $\mathbf{c}$ is **additive:** $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \dots$

Addition doesn't multiply the gradient — it can flow back many steps without shrinking exponentially (as long as $\mathbf{f}_t \approx 1$).

Full LSTM cell diagram



Forget | Input | Candidate | Output

Part 5 — GRU and Practical Architectures

GRU — Gated Recurrent Unit

Cho et al. (2014) — simplified LSTM with only **2 gates** and **no separate cell**.

Reset gate

$$\mathbf{r}_t = \sigma(W_r\,[\mathbf{h}_{t-1}; \mathbf{x}_t])$$

Controls how much of the previous state influences the candidate.

Update gate

$$\mathbf{z}_t = \sigma(W_z\,[\mathbf{h}_{t-1}; \mathbf{x}_t])$$

Candidate + final update

$$\tilde{\mathbf{h}}_t = \tanh(W\,[\mathbf{r}_t \odot \mathbf{h}_{t-1}; \mathbf{x}_t])$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$

GRU vs LSTM

	LSTM	**GRU**
Gates	3 (f, i, o)	2 (r, z)
State	$\mathbf{h}_t$ + $\mathbf{c}_t$	only $\mathbf{h}_t$
Parameters	More	Fewer
Training	Slower	Faster
Performance	Slightly better for very long sequences	Comparable on most tasks

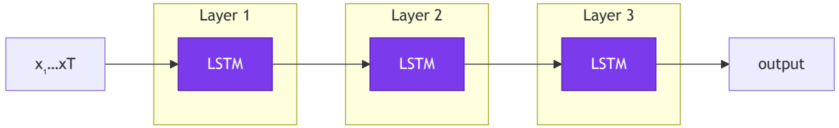
Practical rule: start with GRU if sequences are not extremely long or the dataset is small. Use LSTM when you need very long-term

Bidirectional RNN

Stacked RNNs and Dropout

Stacking

The output of one RNN layer feeds the next — each layer learns more abstract representations.



2–4 layers are common; more than that rarely helps and increases the risk of overfitting.

Dropout in RNNs

Standard dropout between layers works, but **not** inside the recurrent cell (it would destroy memory).

Variational dropout (Gal & Ghahramani, 2016): applies the **same mask** at all time steps.

```
# PyTorch — dropout between LSTM layers
nn.LSTM(
    input_size=128,
    hidden_size=256,
    num_layers=3,
    dropout=0.3,      # applied between layers
    bidirectional=False
)
```

The `dropout` argument of `nn.LSTM` is only applied between intermediate layers — the last layer has no automatic dropout.

Part 6 — Code and Applications

`nn.LSTM` and `nn.GRU` — PyTorch

```python
import torch.nn as nn

x = torch.randn(2, 10, 64)  # (batch, seq, feat)

lstm = nn.LSTM(input_size=64, hidden_size=128,
               num_layers=2, batch_first=True,
               dropout=0.2, bidirectional=False)

h0 = torch.zeros(2, 2, 128)  # (layers, batch, hidden)
c0 = torch.zeros(2, 2, 128)

output, (hn, cn) = lstm(x, (h0, c0))
# output: (2, 10, 128) — all h_t
# hn/cn:  (2, 2, 128)  — last state per layer
```

```python
# GRU — same interface, no c
gru = nn.GRU(
    input_size=64,
    hidden_size=128,
    num_layers=2,
    batch_first=True,
    bidirectional=True  # hidden_size × 2 in output
)

output, hn = gru(x)
# output: (2, 10, 256) — bidirectional → ×2
# hn:     (4, 2, 128)  — 2 layers × 2 directions

# Classification: use last token
last_hidden = output[:, -1, :]  # (2, 256)

# Or mean over the sequence
mean_hidden = output.mean(dim=1) # (2, 256)
```

With `batch_first=False` (default): format `(seq, batch, feat)`. I recommend always using `batch_first=True` for consistency with

Sentiment classification — PyTorch

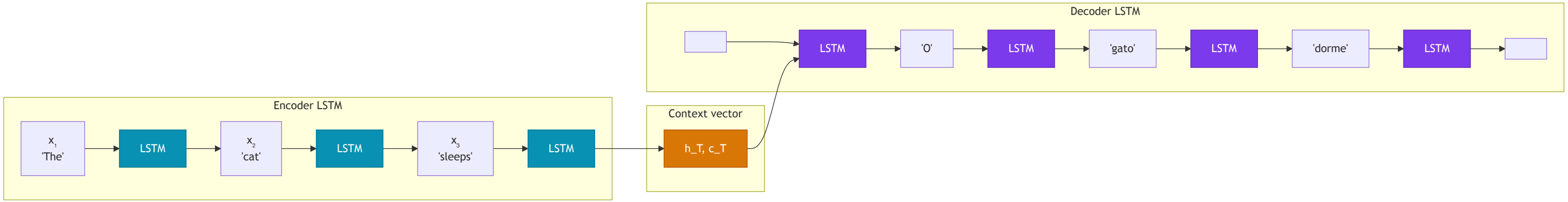
```
class SentimentLSTM(nn.Module):
    def __init__(self, vocab_size, embed_dim, hidden_dim, num_layers, num_classes):
        super().__init__()
        self.embed   = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.lstm    = nn.LSTM(embed_dim, hidden_dim, num_layers,
                               batch_first=True, dropout=0.3)
        self.dropout = nn.Dropout(0.3)
        self.fc      = nn.Linear(hidden_dim, num_classes)

    def forward(self, x):                    # x: (batch, seq_len)
        emb = self.embed(x)                  # (batch, seq_len, embed_dim)
        out, (hn, _) = self.lstm(emb)        # out: (batch, seq_len, hidden)
        # Use last hidden state of last layer
        last = hn[-1]                        # (batch, hidden_dim)
        last = self.dropout(last)
        return self.fc(last)                 # (batch, num_classes) — logits

model     = SentimentLSTM(10_000, 128, 256, 2, 2)
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)

for tokens, labels in train_loader:
    logits = model(tokens)
    loss   = criterion(logits, labels)
```

LSTM in Keras

```python
import tensorflow as tf
from tensorflow import keras

# Full pipeline
vectorizer = keras.layers.TextVectorization(
    max_tokens=10_000,
    output_sequence_length=128
)
vectorizer.adapt(train_texts)

model = keras.Sequential([
    keras.Input(shape=(1,), dtype='string'),
    vectorizer,
    keras.layers.Embedding(10_000, 128, mask_zero=True),
    # Stacked bidirectional LSTM
    keras.layers.Bidirectional(
        keras.layers.LSTM(128, return_sequences=True,
                          dropout=0.2, recurrent_dropout=0
    ),
    keras.layers.Bidirectional(
        keras.layers.LSTM(64, dropout=0.2)
    ),
    keras.layers.Dense(64, activation='relu'),
    keras.layers.Dropout(0.3),
```

```python
model.compile(
    loss='binary_crossentropy',
    optimizer=keras.optimizers.Adam(1e-3),
    metrics=['accuracy']
)

# Train
history = model.fit(
    train_texts, train_labels,
    validation_split=0.2,
    epochs=10, batch_size=64
)
```

Application: sequence-to-sequence (seq2seq)



Encoder: processes the input sequence, compresses it into a context vector (h_T, c_T).

Decoder: initialized with (h_T, c_T), generates the output token by token autoregressively.

Limitation of vanilla seq2seq: the context vector is a bottleneck — compressing an entire sequence into a single vector loses information. The solution is the **attention mechanism** (→ Lecture 6: Transformers).

Comparison: RNN, LSTM, GRU

	Vanilla RNN	LSTM	GRU
Memory	State $\mathbf{h}_t$	$\mathbf{h}_t$ + $\mathbf{c}_t$	only $\mathbf{h}_t$
Gates	—	3 (f, i, o)	2 (r, z)
Parameters	Baseline	$4\times$ more than RNN	$3\times$ more than RNN
Long dependencies	❌ Vanishing	✅ Good	✅ Good
Speed	Fast	Slower	Intermediate
Typical use	Quick baseline	NLP standard until 2018	Efficient alternative

In 2017–2018, LSTMs were the state of the art in translation, summarization and speech recognition. With the arrival of **Transformers** (Vaswani et al., 2017), they were gradually superseded — but RNNs/LSTMs remain useful for low-latency inference, memory-constrained devices and time-

Summary

The problem

- Sequential data has order and variable length
- Dense networks: no order, no weight sharing
- CNNs: limited receptive field

Vanilla RNN

- $\mathbf{h}_t = \tanh(W_h \mathbf{h}_{t-1} + W_x \mathbf{x}_t)$
- Weights shared across all steps
- Suffers from vanishing gradient

BPTT and gradients

- Gradient flows back through chained multiplications
- Vanishing: $\gamma^T \to 0$ with $\gamma < 1$
- Exploding: gradient clipping

LSTM

- Cell $\mathbf{c}_t$ + gates f, i, o
- Additive path: solves vanishing
- $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$

GRU and practices

- 2 gates, more efficient than LSTM
- Bidirectional: past + future context
- Stacking + variational dropout

Next lecture

- Context vector bottleneck
- Attention mechanism
- Full Transformer architecture

Next lecture

Lecture 6 — Transformers

- Seq2seq bottleneck as motivation
- Self-attention mechanism
- Scaled Dot-Product Attention and Multi-Head
- Positional encoding
- Full architecture (encoder + decoder)
- BERT and GPT: fine-tuning and generation

For this week

- Notebook `lec05-rnn.ipynb`:
  - Sentiment classification with LSTM (IMDB)
  - RNN vs LSTM vs GRU comparison
  - Learning curves and gradient analysis
  - Hidden state visualization with t-SNE

Thank you! Questions?

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